

Asymp. Symm:

Global $\rightarrow$ Noether current J^μ (Conserved $\partial_\mu J^\mu$)

$$Q = \int_\Sigma J^\mu d^{d-1}x$$

$$\partial_\mu J^\mu = 0$$

$$\tilde{J}^\mu = J^\mu + \partial_\nu K^{\mu\nu}$$

$\hookrightarrow$ No ext. fluxes

$$\tilde{J} = J + dK$$

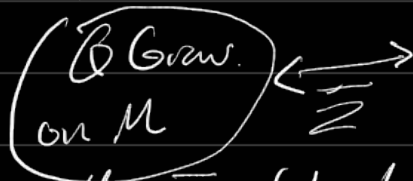
Gauge "Symmetries": $J = \underset{0}{S} + dK$

$$Q = \int_\Sigma J = \int_\Sigma K$$

Q : conserved, $\oint dK = 0$

fields theory on ∂M

Holography:



$AdS_{d+1} \equiv CFT_d$

Asymp symm $\mathcal{M} \equiv$ global symm of dual theory.

* Noether Global:

$$1) \delta S = 0 \quad \varepsilon \rightarrow \varepsilon(x): \quad \partial_\varepsilon S = \int J^\mu \partial_\mu \varepsilon(x)$$

on-shell $\delta S = 0$, $\varepsilon(x)$: compact supp.

$$\int \varepsilon(x) \partial_\mu J^\mu$$

$$\delta S = \int E(x) \delta \phi + \partial_\mu \Theta^\mu(\delta \phi)$$

$\hookrightarrow$ E.O.M.

$$\delta_{\varepsilon=\text{const}} S = \int \partial_\mu M^\mu_\varepsilon$$

$$\text{ex)} \quad x^\mu \rightarrow x^\mu + a^\mu$$

$$M^\mu = a^\mu \mathcal{L} \quad | \quad J^\mu = \Theta^\mu(\delta_\varepsilon \phi) - M^\mu_\varepsilon$$

proof:

$$\mathcal{L}(\phi, \Delta x + a) = \mathcal{L}(\phi, x) + a^\mu \partial_\mu \mathcal{L} + \dots$$

$$= \mathcal{L}(\phi, \Delta x) + a^\mu \frac{\partial}{\partial \Delta x^\mu} \mathcal{L} + \dots$$

$$\text{so that } J^\mu = \varepsilon^\mu \mathcal{L} \quad \text{and}$$

$$\partial_\mu (a^\mu \mathcal{L})$$

$$\partial_\mu J^\mu = -E(\phi) \delta_\varepsilon \phi$$

$$\hookrightarrow J^\mu = \Theta^\mu(\delta_\varepsilon \phi) - M^\mu_\varepsilon$$

$$\partial_\mu J^\mu = \partial_\mu \Theta^\mu - \partial_\mu M^\mu_\varepsilon$$

$$\left(\text{on-shell } \partial_\mu M^\mu_\varepsilon = 0 \right.$$

$$\left. \hookrightarrow E(\phi) \delta \phi + \partial_\mu \Theta^\mu(\delta \phi) = 0 \right)$$

$$\hookrightarrow \boxed{\partial_\mu \mathcal{J}^\mu = -E(\phi) \delta \phi}$$

ex: $\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^*$

i) $\phi \rightarrow e^{i\varepsilon} \phi$ $\mathcal{J}^\mu$

ii) $x^\mu \rightarrow x^\mu + a^\mu$

$$\rightarrow (i) \delta \mathcal{L} = - \int \delta \phi \square \phi^* - \int \square \phi \delta \phi^*$$

$$\delta \phi = i\varepsilon \phi$$

$$\hookrightarrow \delta \phi^* = -i\varepsilon \phi^*$$

$$\begin{aligned} \delta \mathcal{L} &= - \int i\varepsilon \phi \square \phi^* + \int \square \phi i\varepsilon \phi^* \\ &= \int i\partial_\mu \varepsilon (\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi) \end{aligned}$$

$$\therefore \mathcal{J}^\mu = i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi)$$

$\rightarrow x^\mu \rightarrow x^\mu + a^\mu$

$$\mathcal{L}(\phi(x'), x') = \mathcal{L}(\phi) + \delta \phi \frac{\delta \mathcal{L}}{\delta \phi} + \dots$$

$$\delta \phi = \delta x^\mu \partial_\mu \phi \quad \frac{\delta}{\delta \phi} = \frac{\delta x^\sigma}{\delta \phi} \frac{\partial}{\partial x^\sigma}$$

$$\Rightarrow \mathcal{L}(\phi'(x')) = \mathcal{L}(\phi) + \delta x^\mu \left(\frac{\delta \mathcal{L}}{\delta x^\mu} \right) \left(\frac{\delta x^\sigma}{\delta \phi} \right) \partial_\sigma$$

$$\sim \mathcal{L}(\phi) + \delta x^\mu \delta_\mu^\sigma \partial_\sigma \mathcal{L} + \dots$$

$$\therefore \delta \mathcal{L} = \delta x^\mu \partial_\mu \mathcal{L} + \dots \quad \delta x^\mu = a^\mu$$

$$\delta S = \int \delta x^\mu \partial_\mu \mathcal{L} + \dots = \int a^\mu \partial_\mu \mathcal{L} = \int \partial_\mu (a^\mu \mathcal{L})$$

$$\mathcal{J}^\mu = -a^\mu \mathcal{L} + \text{E.O.M terms}$$

local symmetry: $\delta_\varepsilon S = 0$ off shell

$$\delta_\varepsilon S = \int E(\phi) \delta \phi + \nabla_\mu \Theta^\mu(\delta_\varepsilon \phi)$$

$$\delta_\varepsilon \phi = f(\phi) \cdot \varepsilon(x) + f^\mu \partial_\mu \varepsilon(x)$$

ε : compact support. $\Rightarrow \nabla_\mu \Theta^\mu \rightarrow 0$

$$\delta_\varepsilon S = \int E(\phi) (f(\phi) \varepsilon(x) + f^\mu \partial_\mu \varepsilon)$$

$$= \int \underbrace{(E(\phi) f(\phi) - \partial_\mu (E(\phi) f^\mu(\phi)))}_0 \varepsilon(x) \quad (\text{off-shell})$$

$$E(\phi) f(\phi) = \partial_\mu (f^\mu(\phi) E(\phi))$$

$$\rightarrow \text{Eq. } \delta t_2 = \int [f E - f^M \partial_\mu E] + \underbrace{\partial_\mu S^M(E)}_{\text{Boundary terms}}$$

S^M varies on-shell

on-shell varying Noether current (for local symm)

Maxwell: $A_\mu \rightarrow A_\mu + \partial_\mu \epsilon$

$$\begin{aligned} \delta S &= -\frac{1}{4} \int d^4x \, F^{\mu\nu} \partial_\mu \delta A_\nu \\ &= \int d^4x \, \partial_\mu F^{\mu\nu} \delta A_\nu = \int d^4x \, \partial_\mu F^{\mu\nu} \partial_\nu \epsilon(x) \\ &\quad - \underbrace{\text{Boundary} \rightarrow \int \partial_\mu (F^{\mu\nu} \delta A_\nu)}_{\Theta^\mu} \end{aligned}$$

$$\underbrace{\partial_\mu F^{\mu\nu}}_{E^\nu} \delta A_\nu = E^\nu \partial_\nu \epsilon = \underbrace{\partial_\nu (E^\nu \epsilon(x))}_{S^\nu} - \underbrace{\epsilon(x) \partial_\nu E^\nu}$$

$E^\nu \hookrightarrow E, QM.$

$$S^\nu = \partial_\mu F^{\mu\nu} \epsilon(x) \quad []$$

$$\bullet S^M = -J^M - \partial_\nu Q^{\mu\nu} \leftarrow \text{Boundary}$$

$$\Theta^\mu = -F^{\mu\nu} \delta A_\nu = F^{\mu\nu} \partial_\nu \epsilon(x), \quad M^\mu = 0$$

$$J^\mu = -F^{\mu\nu} \partial_\nu \epsilon(x) = -\partial_\nu (\epsilon(x) F^{\mu\nu}) + \underbrace{\epsilon(x) \partial_\nu F^{\mu\nu}}_{-S^\mu}$$

$$Q^{\mu\nu} = \epsilon(x) F^{\mu\nu}$$

$$Q = \int_{\partial \Sigma} \epsilon(x) \epsilon_{\alpha\beta\mu\nu} F^{\mu\nu}$$

